Forage-Citi-MQA_Coffee-Contract-Pricing

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Using the provided market data, apply the cost of carry model, the Black-Scholes model, and Monte Carlo simulations to determine the fair price of a coffee option contract. Analyze how changes in supply, demand, weather, and geopolitical factors might impact your pricing.

Given:

Current spot price (S_t): $1.20 per pound

Risk-free rate (r): 2% per annum

Storage cost (d): 1% per annum

Time to maturity (T): 6 months (0.5 years)

Strike price (X): $1.25

Volatility (σ): 25%

Applying the Models

Cost of Carry Model

Project: Calculate the futures price for a coffee contract maturing in six months using the current spot price, the risk-free interest rate, and the estimated storage costs.

Steps:

1. Gather Data:

- Spot Price: Find the current spot price of coffee. You can obtain this from financial news websites like Bloomberg, Reuters, or commodity-specific sites like the International Coffee Organization (ICO).
- Risk-Free Rate: Use the current yield on a six-month US Treasury bill as a proxy for the risk-free rate. This data can be found on financial websites like the US Department of the Treasury or FRED (Federal Reserve Economic Data).
- Storage Costs: Estimate storage costs as a percentage of the spot price. This information might be available in commodity market reports or industry publications.

2. Calculate Futures Price:

Use the cost of carry formula:

$$F_t = S_t \cdot e^{(r+d)T}$$

Where:

- F_t: Futures price
- S_t: Spot price
- r: Risk-free rate
- d: Storage cost
- T: Time to maturity (0.5 years for six months)

```r
# Given values

S_t = 1.20 # Spot Price in dollars
r = 0.02   # Risk-free rate (2%)
d = 0.01   # Storage Cost (1%)
T = 0.5    # Time to maturity in years

# Calculating futures price

F_t = S_t * exp((r+d)*T)

cat(sprintf("The fair price of the coffee futures contract is $%.3f\n", F_t))

## The fair price of the coffee futures contract is $1.218
```

Black-Scholes Model

Project: Price a call option on a coffee futures contract using the current spot price, strike price, risk-free rate, time to maturity, and volatility.

Steps:

1. Gather Data:

- Spot Price: Use the same spot price data as for the futures contract.
- Strike Price: Choose a strike price based on market conditions and option contracts available on exchanges like the ICE (Intercontinental Exchange).
- Risk-Free Rate: Use the same risk-free rate as for the futures contract.
- Time to Maturity: Specify the time to maturity (e.g., six months).
- Volatility: Calculate or find the historical volatility of coffee prices. This data can be derived from historical price data available on financial websites or through databases like Yahoo Finance.

2. Calculate Option Price:

Use the Black-Scholes formula:

$$C = S_0 N(d_1) - X \cdot e^{-rT} N(d_2)$$

$$d_1 = \frac{ln(S_0/X) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Where:

- C: Call option price
- S_0: Spot price
- X: Strike price
- r: Risk-free rate
- T: Time to maturity
- σ: Volatility

```r
# Given values
S_0 = 1.20 # Spot Price in dollars
r = 0.02   # Risk-free rate (2%)
d = 0.01   # Storage Cost (1%)
T = 0.5    # Time to maturity in years
X = 1.25   # Strike price in dollars
sigma = 0.25    # Volatility (25%)

# Calculating d1 and d2

d_1 = (log(S_0/X)+(r+ 0.5*sigma^2*T))/ sigma* sqrt(T)

d_2 = d_1 - sigma* sqrt(T)

# Calculating call option price using Black-Scholes formula
C = S_0* pnorm(d_1) - X*exp(-r*T)*pnorm(d_2)

C

## [1] 0.06814236

cat(sprintf("The price of the call option is $%.3f", C))

## The price of the call option is $0.068
```

Monte Carlo Simulation

Project: Run simulations to forecast the price of coffee futures under different market conditions. Evaluate the impact of varying supply, demand, and weather scenarios.

Steps:

1. Gather Data:

- Initial Price: Use the same spot price data.
- Risk-Free Rate: Use the same risk-free rate.
- Volatility: Use the same volatility data.
- Simulated Period: Define the time period for the simulation (e.g., six months).

2. Run Simulations:

- Use Monte Carlo simulations to generate numerous possible future price paths based on the inputs.
- Model the stochastic process of coffee prices using Geometric Brownian Motion (GBM).

```r
# Given Values
S_0 = 1.20   # Spot price in dollars
r   = 0.02   # Risk-free rate (2%)
sigma = 0.25 # Volatility (25%)
T   = 0.5    # Time to maturity in years
num_sims = 10000

# Running Monte Carlo Simulation
set.seed(123) # For Reproducibility

simulated_price = S_0 * exp((r-0.5*sigma^2)*T +
sigma*sqrt(T)*rnorm(num_sims))

# Calculating the average simulated price
avg_simulated_price= mean(simulated_price)

cat(sprintf("The average simulated price of the coffee futures contract is
$%.3f", avg_simulated_price))

## The average simulated price of the coffee futures contract is $1.212
```